RIVER VALLEY HIGH SCHOOL

2026 Preliminary Examination
General Certificate of Education Advanced Level
Higher 2

MATHEMATICS 9758/02

Paper 2 **3 hours**

Additional Materials: Writing paper
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.
Write your answers on separate writing paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
You must show all necessary working clearly.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 100.

This document consists of 8 printed pages.

Practice copy rebuilt from the RVHS 2026 mark scheme: questions reproduced exactly, solutions removed.

Section A: Pure Mathematics [40 marks]

1 The curve C has parametric equations

$$x = t + \frac{4}{t}, \quad y = -\frac{2}{t}, \quad 0 < t < 2.$$

(a) Show $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{t^2 - 4}$. Hence find the value of $\frac{\mathrm{d}y}{\mathrm{d}x}$ as $t \to 0$. [3]

(b) The equation of an asymptote of C is of the form $y = kx$, where k is a real constant. Find the value of k. [1]

(c) Sketch C, indicating clearly the coordinates of the endpoint of the curve and the equations of any asymptotes. [2]

2 A start up company is launching a project to capture airborne dust particles using two different filtration models A and B, with its efficiency measured over time.

Model A is programmed such that the amount of dust particles captured in its first month of operation is 8 units and increases by a fixed amount of d units each month. In its first 12 months of operation, Model A captured a total of 360 units of dust particles.

(a) Find the amount of dust particles captured by Model A in its 20th month of operation. [3]

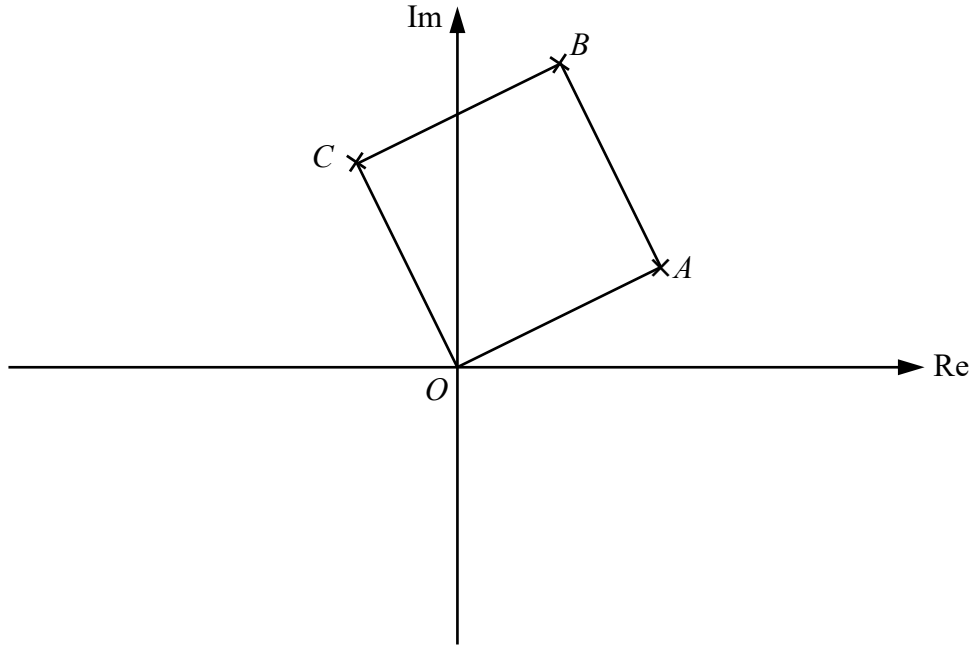
Model B uses a depleting chemical catalyst to capture dust particles. The amount of dust particles captured in the first month is a units. For each subsequent month, the amount of dust particles captured is r times that of the previous month. Over its lifespan, Model B can capture a theoretical maximum of 1000 units of dust particles. In the first two months, Model B captured a total of 640 units of dust particles.

(b) Find the values of a and r. [4]

The startup company will decommission Model B once its total amount of dust particles captured reaches at least 99% of its maximum capacity over its lifespan.

(c) Find the minimum number of months that Model B will be deployed before it is decommissioned. [3]

3



The point A on the Argand diagram represents the complex number z, where $|z| = 3$ and $\arg(z) = \frac{\pi}{7}$. Points B and C are on the diagram such that $OABC$ is a square.

(a) **(i)** The points B and C represent complex numbers v and w respectively. Find v and w in terms of z. [2]

(ii) State the exact values of $|v|$ and $\arg(v)$. [2]

Points D and E represent the complex numbers $-\mathrm{i}z^*$ and $2\,\mathrm{Re}(z)$ respectively.

(b) **(i)** On the same Argand diagram in the Printed Answer Booklet, plot the points D and E. Show clearly the geometrical relationship between the points C and D, and between points A and E. [2]

(ii) Show that the complex number representing $\overrightarrow{DB}$ is $z + \mathrm{i}z + \mathrm{i}z^*$. [1]

(iii) Show that BD is perpendicular to CE. [2]

4 **(a)** Find $\int \frac{1-x^2}{(1+x^2)^2}\,\mathrm{d}x$ using the substitution $x = \tan u$. [6]

The curves C_1 and C_2 have equations $y = \frac{2}{(1+x^2)^2}$ and $y = \frac{1}{1+x^2}$ respectively.

(b) Sketch C_1 and C_2 on the same graph, giving the coordinates of any points where C_1 and C_2 intersect and the equations of any asymptotes. [3]

The finite region R is bounded by C_1 and C_2.

(c) Using the result in part **(a)**, find the area of the region R. [3]

(d) Find the volume of the solid generated when the region R is rotated 2π radians about the x-axis. [3]

Section B: Statistics [60 marks]

5 The probability distribution of a discrete random variable X is given as follows:

x	–2	0	2	4
$\mathrm{P}(X = x)$	p	0.2	q	0.3

Given $\mathrm{E}(X) = 1.2$,

(a) Find the values of p and q. [3]

(b) Find the value of $\mathrm{Var}(X)$. [1]

6 A consumer wants to investigate the average delivery timing by recording 50 timings of food delivery to his house for dinner over 6 months.

(a) In context of the question, give a reason why this method of sampling is not suitable. [1]

A delivery platform's auditor collects the timings from a random sample of 50 food deliveries. The food delivery timings, denoted by x minutes, of the random sample are summarised below.

$$\sum(x-30) = -66.5, \qquad \sum(x-30)^2 = 586.0763$$

(b) Find the unbiased estimates for the population mean and variance. [2]

The platform's auditor chooses another random sample of 45 timings.

(c) Use your estimates from part **(b)** to find the probability that in a randomly chosen sample of 45 delivery timings, the mean delivery timing is between 28 minutes and 35 minutes. [3]

7 The following table shows the number of charge cycles, n, undergone by a battery, and its capacity, as a percentage of its original capacity, p %.

n	100	150	200	250	300	350	400	450
p	96	90	85	83	80	79	77	76

(a) Sketch a scatter diagram of the data, labelling the axes. Explain why the scatter diagram is consistent with a model of the form $p = a + \frac{b}{n}$, where a and b are constants. [2]

(b) For the model of the form $p = a + \frac{b}{n}$, calculate the product moment correlation coefficient and the constants a and b. [2]

(c) Using the model in part **(b)**, estimate the capacity, as a percentage of its original capacity, when the battery undergoes 365 charge cycles. Comment on the reliability of the estimate. [2]

For the model $p = \mathrm{f}(n)$, the residual for a point (c, d) is $d - \mathrm{f}(c)$.

Let S be the sum of the squares of residuals for the model $p = \mathrm{f}(n)$.

For the values of a and b found in part **(b)**,

(d) **(i)** find S when $\mathrm{f}(n) = a + \frac{b}{n}$, and [1]

(ii) state the range of S when $\mathrm{f}(n) = a + \frac{b}{dn}$, where d is non-zero real constant. [1]

8 White truffles are harvested from the forest. According to harvest records, the distribution of the mass of a white truffle has mean 20 grams and standard deviation 3.5 grams.

It is known that climate change affects the mass of white truffles. To investigate these effects, Lucia collects a random sample of 32 white truffles with a mean mass of 19 grams. She decides to test, at the 5% level of significance, whether the mean mass of white truffles has decreased.

(a) State hypotheses for Lucia's test, defining any parameter that you use. [2]

(b) State one assumption Lucia makes to carry out her test. [1]

(c) Carry out Lucia's test at the 5% level of significance. Show the values you use to carry out the test and give your conclusion in the context of the question. [3]

Jason also wishes to investigate whether the mean mass of white truffles has decreased. He collects another random sample of n white truffles, where $n \geq 30$, with a mean mass of k grams.
Past records revealed that across many harvest expeditions that collected n truffles, 98% of them harvested truffles with a mean mass of more than k grams.

(d) State the meaning of p-value in context of Jason's test. [1]

(e) Carry out Jason's test at the 2.5% level of significance. Show the values you use to carry out the test and give your conclusion in the context of the question. [2]

9 **(a)** Events A and B are such that $\mathrm{P}(B' \mid A') = \frac{4}{5}$ and $\mathrm{P}(A) + 5\mathrm{P}(B) = \frac{5}{2}$. Find $\mathrm{P}(A \cap B)$. [4]

(b) A school organises a camp for 6 seniors and 5 juniors.

(i) For the opening ceremony, the 11 students are arranged in a straight line for a photograph. Find the number of possible arrangements if the students standing at each end of the line are not both seniors. [2]

(ii) For a discussion session, the 11 students are seated at a round table. Find the number of possible arrangements if no 2 juniors are seated together. [2]

(iii) For an activity, the 11 students are grouped into a group of 5 and a group of 6. Find the number of ways to group the students if each group has at least 2 juniors. [2]

10 A robot is programmed to move along the x-axis on the x-y plane. It starts at the origin $O(0,0)$.

The robot receives a signal to either:

- move one unit to the right with a probability of 0.3; or
- move one unit to the left, with a probability of 0.7.

The signals are independent of each other.

The robot receives 10 signals. The random variable R denotes the number of times the robot moves to the right.

(a) Find the probability the robot moves exactly 3 times to the right. [1]

(b) Find the probability the robot moves more than 3 but less than 8 times to the right. [2]

The random variable X denotes the x-coordinate of the final position of the robot after the 10 signals.

For example, if the robot moves exactly 8 times to the right (and so, 2 times to the left) it will end up at the final position with x-coordinate 6.

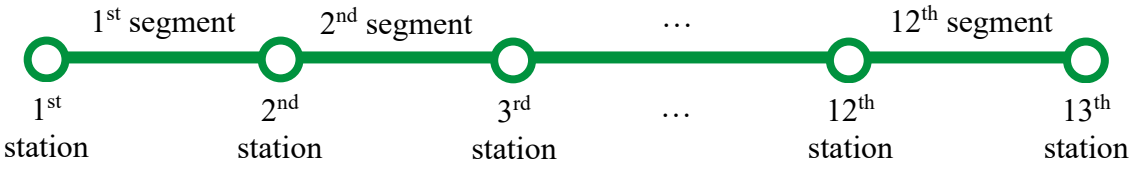
(c) **(i)** If the robot moves exactly r times to the right, show that x-coordinate of the final position of the robot is $2r-10$. [1]

(ii) Find $\mathrm{P}(X=-4)$. [1]

(iii) Find the mean and variance of X. [3]

(d) In a separate experiment involving n independent signals, the probability that the robot moves to the right more than 4 times exceeds 0.2. Find the least possible value of n. [2]

11 Mr Lim travels to work by train every weekday morning. His journey begins at the 1st station and ends at 13th station. The train passes through 12 segments, where each segment is the distance between two consecutive stations. The following diagram illustrates his journey.



Mr Lim leaves the 1st station at 8.25 am every weekday. If Mr Lim arrives at the final station at or before 9 am, he is considered to be early. Otherwise, he is considered to be late.

On fine days, the travel time for each train segment is independent and normally distributed with mean 2.5 minutes and standard deviation 0.8 minutes.

(a) Show that, on a randomly chosen fine day, the probability that he arrives early for work is 0.9644, correct to 4 decimal places. [2]

(b) Over 75 fine days, find the expected number of times Mr Lim arrives early for work. [1]

On rainy days, the train takes 20% longer to travel across each train segment.

(c) Show that, on a randomly chosen rainy day, the probability that he is late for work is 0.6182, correct to 4 decimal places. [3]

(d) On rainy days, Mr Lim wishes to change his departure time so that he is early for work on at least 95% of the rainy days. Find the latest time, to the nearest minute, that he should leave the first station on a rainy day. [3]

It is known that there is a probability of p that it rains on a weekday morning.
Regardless of the weather, it is also known that when Mr Lim leaves the first station at 8.25 am, he is early for work 80% of the time.

(e) One day, Mr Lim is late for work. Find the probability it was a rainy day. [4]

END OF PAPER